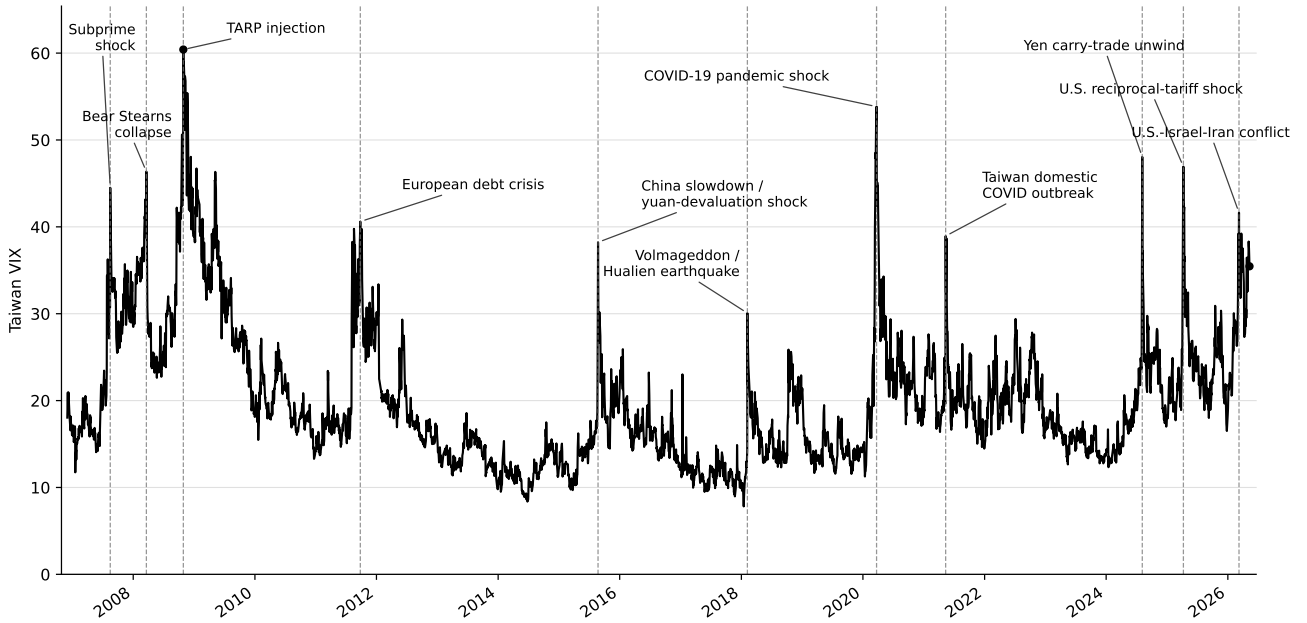


1 Research Project Background

1.1 Motivation and Taiwan Context

Taiwan provides a particularly important setting for studying external uncertainty transmission. It is deeply connected to the United States through financial markets, monetary conditions, and technology demand, and it is closely linked to China through trade, production networks, and cross-strait risk. At the same time, Taiwan occupies a central position in global semiconductor supply chains. These features make Taiwan highly exposed to external uncertainty. As Figure 1 shows, Taiwan's option-implied volatility rises sharply around several major global and regional uncertainty episodes.

Figure 1: Taiwan VIX and major uncertainty episodes



Exposure to these external sources makes identification difficult because the same external disturbance can reach Taiwan through several channels at once. A U.S. monetary tightening, for example, may affect Taiwan through capital flows and exchange rates, but it may also affect Taiwan through export demand and firms' production expectations. Similarly, a China slowdown may operate through trade and supply-chain linkages in one episode and through financial-market sentiment in another. The same external shock may therefore look financial in one episode, macroeconomic in another, and mixed in periods of severe stress.

This project therefore asks how external uncertainty shocks from the United States, China, and the global economy operate in Taiwan, and which structural friction explains their macroeconomic effects. It answers this question by moving from *source*, to *channel*, to *mechanism*. Year 1 identifies the source and reduced-form transmission channel by estimating Taiwan's latent uncertainty factors and classifying external drivers based on their time-varying association with these factors. However, reduced-form evidence alone cannot distinguish between competing economic theories. For instance, a financial-channel response could be driven by intertemporal collateral constraints or static working-capital frictions.

Year 2 addresses this challenge by providing a *structural mapping*: it derives a projection

framework that converts second-moment uncertainty risk corrections into first-moment Business Cycle Accounting (BCA) wedges. This theoretical bridge allows us to define unique “diagnostic signatures” for each friction—predicting, for example, that collateral shocks should load primarily on the investment wedge while working-capital frictions should load on the after-tax labor wedge. Finally, Year 3 validates these mechanisms empirically by constructing Taiwan’s BCA wedges and computing MCMC-integrated wedge impulse response profiles to test which theoretical signature best matches the empirical response. By integrating these three stages, the project moves beyond simple reduced-form descriptions to provide a structural diagnosis of Taiwan’s external vulnerabilities.

1.2 Open Challenges and Relation to the Literature

A direct identification of how external uncertainty reaches Taiwan must overcome four overlapping difficulties. Each has been engaged separately by prior literature, but no existing study addresses them jointly. The four challenges below organize what is missing and motivate the methodological choices described in the next subsection.

The first challenge is that external and domestic sources of uncertainty are tightly intertwined. A single geopolitical or policy event—an FOMC tightening, a U.S.–China tariff round, or a cross-strait incident—may simultaneously alter global risk sentiment, Taiwan’s domestic political outlook, and expectations about trade, production, and finance. In a small model with limited domestic controls, these domestic movements may be misattributed to external shocks, producing severe omitted-variable bias (Carriero, Clark and Marcellino, 2018). The macro-uncertainty literature has documented the broad effects of uncertainty on investment, hiring, and economic activity (Bloom, 2009, 2014; Baker, Bloom and Davis, 2016; Handley and Limao, 2022), but these studies typically rely on a single proxy index or a small model and cannot separate domestic from external contributions. The closest Taiwan-focused precedent, Huang, Yeh and Chen (2021), constructs a Baker–Davis-style text-based EPU index for Taiwan and uses a Diebold–Yilmaz spillover model to examine linkages with U.S. and Japanese EPU indices, finding that roughly half of Taiwan’s economic uncertainty originates domestically and that U.S. and Japanese spillovers are significant. Even this work, however, delivers a single scalar EPU index and a reduced-form spillover decomposition: it does not separate Taiwan’s domestic uncertainty into macroeconomic and financial components, does not identify whether external uncertainty enters Taiwan through macroeconomic or financial channels, and does not estimate uncertainty jointly with its macroeconomic effects in one model. To close all three of these gaps, the present project builds a unified monthly dataset that combines Taiwan domestic anchors with U.S., Chinese, and global drivers, and estimates two latent uncertainty factors jointly with the VAR dynamics that govern their macroeconomic effects.

The second challenge is that the large information set needed to reduce omitted-variable bias creates its own identification problem. In high-dimensional VAR systems, standard Cholesky identification becomes fragile because the results depend on arbitrary variable orderings. At the same time, the strict block-exogeneity restrictions traditionally imposed in small-open-economy models are too rigid for the present question, because they predetermine how external and domestic variables can interact—a particularly costly assumption when the goal is to learn the operative transmission channel from the data. The order-invariant stochastic volatility in mean vector autoregression (OI-SVMVAR) framework of Davidson, Hou and Koop (2025) resolves the ordering problem and accommodates large information sets without imposing recursive zero restrictions. Its “unclassified-block” mechanism has so far been used to classify ambiguous domestic variables,

for example financial conditions indices that might plausibly belong to either the macroeconomic or financial block. It has not yet been adapted to identify the transmission channel of external shocks into a small open economy. The present project repurposes the unclassified block in exactly that way: external drivers are placed in the unclassified block, and the posterior classification probabilities then reveal whether each external driver co-moves with Taiwan’s macroeconomic or financial uncertainty factor in any given month.

The third challenge concerns how uncertainty is measured and how the estimate is used downstream. Many existing approaches rely either on proxy indicators (such as the VIX or a text-based EPU index) or on a two-step design that first estimates uncertainty and then studies its macroeconomic effects in a separate model. Proxy indicators may not capture how uncertainty is experienced inside Taiwan, and two-step procedures treat the first-stage estimate as observed data, discarding estimation uncertainty and inducing measurement-error and model-consistency bias (Carriero, Clark and Marcellino, 2018). Davidson, Hou and Koop (2025) addresses the first-stage problem by estimating uncertainty and its effects jointly. The literature has not, however, propagated first-stage posterior uncertainty into downstream structural-mechanism tests; downstream validation typically uses a point estimate of the latent uncertainty factor and a single posterior-mean impulse response. The present project closes this residual gap with an *MCMC-integrated* wedge impulse response construction: each Year 1 OI-SVMVAR posterior draw delivers its own impulse response path, which is converted into a BCA wedge response through the accounting identities. The resulting credible bands integrate first-stage posterior uncertainty directly into the mechanism test, providing a coherent end-to-end Bayesian inference rather than a hybrid Bayesian-frequentist pipeline.

The fourth challenge is that even a well-identified reduced-form impulse response does not by itself reveal the underlying mechanism. A financial-channel response may reflect a collateral constraint, a time-varying risk premium, a working-capital friction, or several of these at once. The Business Cycle Accounting (BCA) literature (Chari, Kehoe and McGrattan, 2007) provides a useful diagnostic by decomposing observed aggregate fluctuations into wedges that summarize distortions in prototype equilibrium conditions, and small-open-economy extensions add efficiency and external wedges that capture imported-input dependence and exchange-rate movements. BCA has, however, been used primarily as a positive accounting device for ordinary business-cycle dynamics; it has not been connected to uncertainty-driven second-moment risk corrections, and there is no existing wedge-projection framework that converts uncertainty-shock impulse responses into testable wedge signatures. The present project derives such a framework: it shows how collateral and risk-premium frictions should project primarily into the investment wedge, while working-capital frictions should project primarily into the after-tax labor wedge. The reduced-form transmission-channel evidence from Year 1 can then be matched against these structural signatures in Year 3, allowing the project to distinguish competing mechanisms rather than stopping at reduced-form descriptions.

Together these four gaps define the project’s methodological contribution. The next subsection describes the empirical design that addresses them, and the final subsection explains how the design is implemented across the three years.

1.3 Research Design and Main Contributions

The empirical design responds to the four challenges of the previous subsection through three design choices, and yields one project-level methodological contribution plus three year-specific

contributions.

Joint estimation in a large information set. The project uses the order-invariant stochastic volatility in mean vector autoregression (OI-SVMVAR) framework of Davidson, Hou and Koop (2025) to estimate Taiwan’s macroeconomic and financial uncertainty factors jointly with their VAR dynamics. The model allows both latent uncertainty factors to enter the conditional mean of the VAR, so uncertainty directly affects macroeconomic and financial outcomes rather than merely altering their variances. Joint estimation addresses the omitted-variable problem (C1) by accommodating a rich information set covering both domestic anchors and multiple external drivers; the order-invariant identification scheme addresses the large-VAR ordering problem (C2) without requiring restrictive block-exogeneity assumptions; and the single-stage Bayesian estimation removes the measurement-error bias of two-step procedures (C3).

External drivers as an unclassified block. The central adaptation is to repurpose the OI-SVMVAR unclassified-variable mechanism for transmission-channel identification. Taiwan’s clearly real-side variables (industrial production, exports, employment, prices) anchor the domestic macroeconomic uncertainty factor, while Taiwan’s clearly financial variables (equity returns, exchange-rate volatility, credit conditions) anchor the domestic financial uncertainty factor. External drivers—U.S. monetary stance, Chinese real-side cyclical indicators, global financial-market volatility, geopolitical risk, and trade-policy uncertainty—are placed in the unclassified block. The posterior classification probabilities then deliver a data-driven, time-varying measure of which channel each external driver activates in any given month, without requiring the researcher to assume the channel *ex ante*.

Wedge projection and MCMC-integrated validation. Reduced-form classification cannot by itself distinguish competing structural frictions (C4). Year 2 derives a wedge-projection framework that maps uncertainty-related second-moment risk corrections in a nonlinear structural model into Business Cycle Accounting wedges, converting transmission-channel evidence into testable structural predictions: collateral and risk-premium mechanisms should load primarily on the investment wedge, while working-capital mechanisms should load primarily on the after-tax labor wedge. Year 3 then constructs MCMC-integrated wedge impulse response profiles by propagating each OI-SVMVAR posterior draw through the BCA accounting identities, and compares these empirical profiles against the theoretical wedge generalized impulse response functions implied by the candidate mechanisms. By propagating first-stage posterior uncertainty into the structural test rather than relying on a point estimate, this MCMC-integrated design closes the residual two-step bias of C3 and yields a coherent end-to-end inference path.

The project’s central **methodological contribution** is to repurpose the OI-SVMVAR unclassified-block mechanism. Where the original framework uses it to classify ambiguous domestic variables, the present project uses it to identify the transmission channel of external shocks into a small open economy. This reframing turns a domestic-variable classification tool into a structural diagnostic for external-shock transmission, and it makes the channel of transmission a posterior object rather than an *ex ante* assumption.

The project also produces three year-specific contributions. The **first-year contribution** is *empirical identification*: the project constructs a monthly Taiwan external-uncertainty dataset, estimates Taiwan-specific macroeconomic and financial uncertainty factors, and computes time-varying classification probabilities for each external driver. The **second-year contribution** is *theoretical mapping*: the project derives wedge signatures for collateral, risk-premium, and working-capital frictions in a small-open-economy environment, clarifying how each friction should appear in the investment, labor, efficiency, and external wedges. The **third-year contribution** is *em-*

pirical validation: the project constructs Taiwan’s BCA wedges and uses the MCMC-integrated wedge impulse response profiles to test which structural signature best matches the reduced-form transmission patterns identified in Year 1.

1.4 Three-Year Roadmap

The first year produces the empirical foundation. Its main deliverables are (a) a monthly Taiwan external-uncertainty dataset that combines domestic macroeconomic anchors, domestic financial anchors, and external drivers from the United States, China, and the global economy, with variable lists, time spans, and sources documented in the first-year methodology section; (b) the joint posterior distribution of Taiwan’s macroeconomic and financial uncertainty factors from the OI-SVMVAR estimation; (c) time-varying classification probabilities for each external driver, indicating whether it co-moves with Taiwan’s macroeconomic or financial uncertainty in any given month; and (d) impulse responses and forecast error variance decompositions characterizing how each external uncertainty shock reaches Taiwan’s real and financial variables.

The second year produces the theoretical mapping. Its main deliverables are (a) the mathematical derivation of how second-moment risk corrections in a nonlinear structural model project into Business Cycle Accounting wedges; (b) wedge signatures for the three competing frictions—collateral constraints, time-varying risk premia, and working-capital frictions—identifying which BCA wedge each friction loads on; and (c) a characterization of the efficiency and external wedges that arise specifically in a small open economy with imported-input dependence and exchange-rate exposure. The theoretical derivation is intentionally independent of Year 1’s empirical outputs, so the second-year work proceeds in parallel with the first-year estimation rather than waiting for it to complete.

The third year produces the integrated empirical test. Its main deliverables are (a) Taiwan’s BCA wedges constructed from national accounts data; (b) MCMC-integrated wedge impulse response profiles, built by propagating each Year 1 OI-SVMVAR posterior draw through the BCA accounting identities and benchmarked against the theoretical wedge generalized impulse responses from Year 2; (c) a comparative mechanism indicator—an RMSE-based fit criterion with correlation-based robustness checks—ranking the competing frictions against Taiwan’s data; and (d) policy interpretation that connects the winning mechanism to specific instruments, such as asset-price and credit-market policy for collateral and risk-premium channels and short-term liquidity support for working-capital channels. The third-year estimation therefore depends on Year 1’s posterior draws and on Year 2’s wedge-signature hypotheses, and the project schedule places it only after both inputs are available.

Taken together, the three years yield a three-dimensional account of external uncertainty transmission to Taiwan: which source contributes most, which channel carries it, and which structural friction explains its macroeconomic effects. The integrated final deliverable is a source-by-channel-by-mechanism mapping in which each external driver is associated with its dominant transmission channel and its preferred structural interpretation, providing both an empirical diagnostic for monitoring purposes and a model-class prescription for forecasting under uncertainty.

2 Preliminary Progress and Feasibility

2.1 Preliminary Progress

The project is feasible because it builds on substantial preliminary work. For the first-year analysis, I have developed a preliminary raw-data collection pipeline for most variables in the proposed OI-SVMVAR design, and the dataset has moved from a purely conceptual plan to an operational working structure. The current working design considers a baseline monthly panel covering 2003M1 to 2025M12, with a longer price-index robustness sample beginning in 2001M1. At this stage, the candidate baseline contains 52 variables, including 22 Taiwan macroeconomic variables, 8 Taiwan financial variables, and 22 unclassified variables that include external drivers and Taiwan-domestic variables whose transmission channel may vary over time. Ongoing work focuses on finalizing variable definitions, checking cross-source consistency, refining transformation rules, and documenting the reproducibility pipeline.

The raw data collection already covers Taiwan macroeconomic indicators from the Directorate-General of Budget, Accounting and Statistics (DGBAS), Ministry of Economic Affairs (MOEA), Ministry of Finance (MOF), and National Development Council (NDC) web sources; Taiwan financial and credit variables from the Central Bank of the Republic of China (CBC), the Taiwan Stock Exchange (TWSE), Federal Reserve Economic Data (FRED), and housing-price sources; U.S. variables from FRED and PolicyUncertainty.com; China variables from National Bureau of Statistics (NBS) and People’s Bank of China (PBoC) based sources; and global variables from FRED, Bank for International Settlements (BIS), the Iacoviello geopolitical risk dataset, and PolicyUncertainty.com. The remaining first-year data task is therefore not basic data collection, but construction of the final estimation matrix: harmonizing dates, applying transformations, materializing derived variables such as the term spread and exchange-rate volatility, documenting short endpoint gaps, and producing the balanced panel used in the Bayesian estimation.

The most important preliminary progress is on the second-year theoretical component. To ensure the feasibility of the second-year theoretical component, I have developed a preliminary proof-of-concept for the wedge projection mechanism that turns the informal idea that “uncertainty appears as a BCA wedge” into a formal theoretical framework. Because intertemporal financial frictions primarily distort the household Euler equation, their natural BCA counterpart is the investment wedge. The central result is a projection equation for the BCA investment wedge:

$$\hat{\tau}_{x,t} = \omega_{\tau} E_t \hat{\tau}_{x,t+1} + \frac{1}{2} \Phi(\Omega_t),$$

where $\Phi(\Omega_t)$ is the second-order risk correction generated by stochastic volatility in the true nonlinear structural model, ω_{τ} is a forward-looking coefficient related to the steady-state discount factor, capital depreciation, and adjustment-cost curvature, and Ω_t denotes the stochastic-volatility state. Under an AR(1) law of motion for volatility, this equation admits a closed-form solution. This result shows that a second-moment uncertainty signal in a nonlinear Dynamic Stochastic General Equilibrium (DSGE) model can appear as a persistent first-order investment wedge in a BCA prototype.

The preliminary theoretical work also derives distinct wedge signatures for two competing mechanisms. In a Fernández-Villaverde–Guerrón-Quintana-style collateral or risk-premium channel, uncertainty affects the intertemporal Euler equation through second-order risk corrections (Fernández-Villaverde et al., 2011; Fernández-Villaverde and Guerrón-Quintana, 2020). This mech-

anism projects mainly into the investment wedge. In contrast, a working-capital channel affects firms' static labor-demand condition because firms must finance part of their wage bill before production. This mechanism projects mainly into the after-tax labor wedge. These two mechanisms therefore generate different empirical predictions.

Building on this baseline, a major task for the second year is to extend the projection argument to third order. This extension is highly nontrivial but important because uncertainty shocks enter higher-order perturbation solutions through volatility innovations and state-dependent terms. The third-order generalization introduces vol-of-vol effects, level-volatility interaction terms, and asymmetric responses to positive and negative uncertainty innovations. These objects provide additional empirical handles for distinguishing structural mechanisms.

The current numerical validation also clarifies the remaining tasks. Preliminary calculations show that the theoretical closed-form impact object should not be compared mechanically with the peak of a long-horizon impulse response. The next revision will compare the full horizon profile, including the impact response, sign changes, and peak response. This issue does not weaken the project. Instead, it provides a clear and manageable task for the second-year theoretical and numerical validation stage.

2.2 Feasibility of the Proposed Plan

The proposed three-year plan is feasible for four reasons.

First, the first-year empirical model is based on an existing and published econometric framework. The project does not require developing a new Bayesian sampler from scratch. The main task is to adapt the OI-SVMVAR framework to the Taiwan setting, construct the dataset, and reinterpret the unclassified-variable mechanism as a transmission-channel identifier for a small open economy.

Second, the data construction is manageable because the main raw-data downloads have already been implemented. The repository now contains source-specific scripts for FRED, CBC, TWSE, DGBAS, the Taiwan statistical portal, BIS, China indicators, and CSV-based uncertainty measures. These scripts have produced the raw series needed for the 52-variable design. The remaining data risks are limited and well-defined: foreign institutional net buying begins after the baseline start, global EPU currently has a short endpoint gap, and the China block requires continued source verification. These issues can be handled through transparent sample trimming, robustness specifications, or reduced China-block benchmarks without changing the research design.

Third, the second-year theoretical component already has a clear foundation. The preliminary report has derived the main projection equation, the closed-form solution under AR(1) volatility, the distinction between investment-wedge and labor-wedge signatures, and the need for a third-order extension. The remaining work is to refine the proofs, standardize notation, improve the numerical validation, and adapt the theoretical presentation to the Taiwan application.

Fourth, the third-year empirical validation uses outputs from the first two years rather than requiring a separate data project. Once the first-year OI-SVMVAR produces posterior draws of the uncertainty factors and the second-year theory produces wedge signatures, the third-year task is to construct Taiwan's BCA wedges, compute MCMC-integrated GIRF profiles, and compare them against theoretically simulated wedge paths using a quantitative RMSE criterion. The design propagates first-year posterior uncertainty directly into the GIRF calculations, which eliminates the generated-regressor bias that would arise from treating the latent factors as observed.

Overall, the project has a clear fallback structure. If the full 52-variable OI-SVMVAR is slow to converge, I will first estimate a reduced benchmark model using the core Taiwan anchors and the most important external drivers. If some interpolated variables affect the results, I will report robustness checks that exclude them. If the empirical wedge responses do not sharply select one mechanism, I will report the relative strength of evidence across mechanisms rather than forcing a binary conclusion. These safeguards make the project ambitious but tractable.

3 The First Year: Methods, Procedures, and Implementation Schedule

3.1 Purpose of the First Year

The first year identifies how external uncertainty shocks enter Taiwan and whether they operate through macroeconomic or financial channels. The empirical task is not simply to construct an uncertainty index. Instead, the goal is to jointly estimate Taiwan’s domestic uncertainty factors, measure the effects of these factors on macroeconomic and financial variables, and classify the transmission channel of external drivers over time.

The first-year analysis has four steps. First, I construct a monthly dataset that combines Taiwan’s domestic macroeconomic variables, domestic financial variables, U.S. indicators, Chinese indicators, global risk measures, and trade-policy uncertainty measures. Second, I estimate an order-invariant stochastic volatility in mean vector autoregression (OI-SVMVAR). Third, I extract two latent domestic uncertainty factors: Taiwan macroeconomic uncertainty and Taiwan financial uncertainty. Fourth, I use the posterior classification probabilities of unclassified variables to identify whether each external driver is transmitted to Taiwan through the macroeconomic or financial channel at each point in time.

This first-year component produces the empirical foundation for the whole project. The extracted uncertainty factors become the key inputs for the third-year BCA validation. The time-varying classification probabilities also provide the first direct evidence on whether U.S., Chinese, and global shocks affect Taiwan through different channels in different episodes.

3.2 Empirical Strategy Overview

The empirical framework follows the OI-SVMVAR model of Davidson, Hou and Koop (2025). Let y_t denote the $N \times 1$ vector of monthly observables. In the baseline application, $N = 52$. The observation equation is

$$y_t = \sum_{\ell=1}^p B_{\ell} y_{t-\ell} + \sum_{j=0}^q A_j^h h_{t-j} + B_0^{-1} \varepsilon_t^y, \quad \varepsilon_t^y \sim N(0, U_t),$$

where

$$h_t = (h_{m,t}, h_{f,t})'$$

contains Taiwan’s macroeconomic uncertainty factor $h_{m,t}$ and financial uncertainty factor $h_{f,t}$. The stochastic-volatility-in-mean term,

$$\sum_{j=0}^q A_j^h h_{t-j},$$

allows uncertainty to affect the conditional mean of the economy. This feature is essential for the project because uncertainty may change investment, employment, asset prices, credit conditions, and exchange rates, not merely the variance of these variables. Appendix A provides the compact model statement, including the log-volatility decomposition, the latent classification states, and the posterior sampling objects.

The contemporaneous impact matrix B_0 is unrestricted apart from standard normalizations. This avoids the arbitrary ordering problem that arises in Cholesky-based VARs. In a large model with many domestic and external variables, no single recursive ordering has a strong economic justification. The order-invariant structure is therefore important for the credibility of the impulse responses, forecast error variance decompositions, and classification probabilities.

The conditional variance matrix U_t is diagonal and is organized into three blocks:

$$U_t = \begin{pmatrix} \Omega_{m,t} & 0 & 0 \\ 0 & \Omega_{u,t} & 0 \\ 0 & 0 & \Omega_{f,t} \end{pmatrix}.$$

The macroeconomic block anchors $h_{m,t}$, the financial block anchors $h_{f,t}$, and the unclassified block is allowed to switch between the two factors. For a pre-classified macroeconomic variable, the common volatility component is $h_{m,t}$. For a pre-classified financial variable, the common volatility component is $h_{f,t}$. For an unclassified variable, the common volatility component depends on a latent state:

$$\omega_{i,t}^u = \eta_{i,t}^u + h_{s_{i,t},t}, \quad s_{i,t} \in \{m, f\}.$$

The state $s_{i,t}$ follows a two-state Markov chain. The key object for this project is the posterior classification probability

$$\pi_{i,t} = P(s_{i,t} = m \mid y_{1:T}).$$

The idiosyncratic term $\eta_{i,t}^u$ captures variable-specific volatility that is not absorbed by the common macroeconomic or financial uncertainty factor. When $\pi_{i,t}$ is close to one, the model assigns unclassified variable i to the macroeconomic uncertainty factor at time t . When $\pi_{i,t}$ is close to zero, the model assigns it to the financial uncertainty factor. For an external variable, this probability is interpreted as a time-varying indicator of whether the variable is more closely associated with Taiwan’s macroeconomic or financial uncertainty factor. The classification probability is therefore a channel-association measure; the direction and quantitative importance of external shocks are evaluated jointly with forecast error variance decompositions and impulse responses.

A natural alternative would be to impose a small-open-economy block-exogeneity restriction, under which external variables can affect Taiwan contemporaneously but Taiwan variables cannot affect external variables. I do not impose this restriction in the baseline model for two reasons. First, block exogeneity would introduce new zero restrictions into the contemporaneous impact matrix and would compromise the order-invariant structure that motivates the OI-SVMVAR. Second, Taiwan’s role in global semiconductor supply chains makes strict contemporaneous zero restrictions less innocuous than in a textbook small open economy. The project instead treats the small-open-economy assumption as an empirical diagnostic. If Taiwan’s uncertainty factors

explain little of the forecast error variance of external variables, the small-open-economy restriction is approximately supported by the data. If they explain a non-negligible share, imposing the restriction *ex ante* would have been misleading.

3.3 Variable Classification and Taiwan Adaptation

The main innovation in the first-year empirical design is the way the unclassified-variable mechanism is used in a small open economy. In the original OI-SVMVAR framework, unclassified variables are domestic variables whose macroeconomic or financial nature is ambiguous. In this project, the same statistical device is repurposed to study external transmission. External drivers are placed in the unclassified block because their transmission channel is precisely the object of interest.

Taiwan’s clearly real-side variables anchor the macroeconomic uncertainty factor. These include industrial production, manufacturing production, export orders, retail sales, food-service sales, exports, imports, price indices, unemployment, employment, and real wages. Taiwan’s clearly financial variables anchor the financial uncertainty factor. These include the overnight interbank rate, the ten-year government bond yield, the term spread, stock-market returns, realized equity volatility, turnover, foreign institutional net buying, and margin buying.

The unclassified block contains three types of variables. The first type is external drivers, including U.S. monetary and real indicators, Chinese real and monetary indicators, global financial and geopolitical risk measures, and trade-policy uncertainty measures. The second type is Taiwan domestic variables whose interpretation can vary over time, such as monetary aggregates, credit aggregates, the exchange rate, the real effective exchange rate, the TAIEX level, housing prices, and short-selling balances. The third type is policy and market variables that can behave as either macroeconomic or financial indicators depending on the episode.

This classification is central to the research design. Pre-classifying the U.S. Federal Funds Rate as financial, or China’s industrial production as macroeconomic, would impose the answer before estimation. In reality, U.S. monetary tightening may affect Taiwan through financial channels such as capital flows and exchange-rate pressure, but it may also affect Taiwan through macroeconomic channels such as export demand and production expectations. Similarly, a slowdown in China may operate through Taiwan’s trade and production networks in one episode, and through asset prices and risk sentiment in another. By placing these variables in the unclassified block, the model lets the data determine the operative channel.

The resulting classification probabilities provide a time-varying narrative of external transmission. For example, if the posterior probability for the U.S. Federal Funds Rate is close to one during a given period, the model indicates that U.S. monetary uncertainty is more closely associated with Taiwan’s macroeconomic uncertainty. If the probability is close to zero, the model indicates that the same external driver is more closely associated with Taiwan’s financial uncertainty. Plotting these probabilities across episodes such as the Global Financial Crisis, the U.S.–China trade war, the COVID-19 shock, and the 2022 Federal Reserve tightening cycle allows the project to compare how Taiwan’s exposure changes over time.

3.4 Data Construction and Transformations

The baseline dataset is a balanced monthly panel covering 2003M1–2025M12. The current design contains 52 variables: 22 macroeconomic variables, 8 financial variables, and 22 unclassified variables. A longer price-index robustness sample covers 2001M1–2025M12 for specifications that do not require the TWSE total-return index. The baseline starts in 2003 because the TWSE total-return index, which provides the preferred equity-return measure, begins in January 2003. This starting point also captures Taiwan’s post-WTO integration period while avoiding the earlier Asian financial crisis as a major structural break.

The monthly frequency is central to the empirical design. Quarterly data would be too coarse to capture short uncertainty episodes associated with policy announcements, tariff negotiations, geopolitical events, and financial-market stress. Daily data would capture financial volatility but would exclude many real-side variables needed to anchor the macroeconomic uncertainty factor. Monthly data therefore provide the appropriate compromise: they preserve enough time-series variation to identify short-lived uncertainty episodes while still allowing the model to include production, trade, employment, prices, and credit variables.

Block	Main variables	Main sources
<i>Taiwan</i>		
Macroeconomic	Industrial production, manufacturing production, export orders, retail sales, exports, imports, business-cycle indicators, prices, unemployment, employment, real wages, working hours, and residential construction permits	DGBAS, MOEA, MOF, NDC
Financial	Overnight interbank rate, 10-year government bond yield, term spread, TAIEX returns, TAIEX realized volatility, turnover, foreign institutional net buying, and margin buying	CBC, TWSE
Unclassified	CBC discount rate, money aggregates, bank loans, consumer loans, TAIEX level, housing prices, exchange rates, exchange-rate volatility, real effective exchange rate, and short-selling balance	CBC, TWSE, FRED, BIS, housing-price sources
<i>External</i>		
Unclassified	U.S. Federal Funds Rate, U.S. industrial production, U.S. credit spread, U.S. EPU, China industrial production, China PPI, China M2, VIX, global geopolitical risk, global EPU, and U.S. trade-policy uncertainty	FRED, NBS/PBoC-based sources, PolicyUncertainty.com, GPR dataset, BIS

The Taiwan macroeconomic block has been downloaded from the DGBAS nstatdb system and the official Taiwan statistical portal, supplemented by MOEA, MOF, and NDC series. It includes

production, trade, prices, labor-market, wage, working-hour, business-cycle, and housing-permit variables. The financial block uses CBC and TWSE data, including interest rates, monetary and credit aggregates, TAIEX total-return and price-index measures, turnover, foreign investor flows, margin buying, and short-selling balances. The external block uses FRED, NBS/PBoC-based China series, BIS, PolicyUncertainty.com, and the Iacoviello geopolitical risk dataset. This block includes U.S. monetary, real, credit, EPU, and trade-policy uncertainty variables; China industrial production, PPI, and M2; VIX; global EPU; global geopolitical risk; and Taiwan’s real effective exchange rate.

All series are transformed before estimation to improve stationarity and comparability. Real-side level variables, price indices, employment, wages, money, and credit variables are expressed as year-on-year percentage changes. Interest-rate levels are differenced when they display persistent nonstationarity, while spreads are retained in levels. Equity returns are monthly log returns. Realized volatility measures are computed from daily returns within each month. Exchange-rate and equity-market levels are retained as log levels only when their level is economically meaningful for the classification exercise. All transformed variables are standardized to have zero mean and unit variance before estimation.

Beyond the standard transformations, I address special data-construction choices directly. For example, Taiwan’s legacy wholesale price index (WPI) ends in 2022M12, while the newer producer price index (PPI) begins in 2021M1. Because the two series overlap from 2021M1 to 2022M12, I have constructed a spliced upstream price-pressure series and retained overlap diagnostics. This series is labeled as an auxiliary upstream price measure rather than as an official continuation of the WPI.

At the current stage, the remaining data issues are narrow rather than fundamental. Foreign institutional net buying begins in 2004M4, so the baseline will either trim this series, treat it as a robustness variable, or use a reduced balanced sample when it is included. Global EPU is currently missing the final month of 2025, which can be handled by waiting for the source update or ending the common sample in 2025M11. The term spread is fully determined by the downloaded 10-year government bond yield and overnight interbank rate and will be materialized in the final dataset-building script. For the China block, the current raw-data pipeline has already obtained industrial production, PPI, and M2 series; the remaining task is source validation and robustness documentation.

3.5 Estimation, Inference, and Main Outputs

The model is estimated using the Bayesian Markov Chain Monte Carlo (MCMC) algorithm developed for the OI-SVMVAR framework. The estimation jointly samples the VAR coefficients, the contemporaneous impact matrix, the common log-volatility factors, idiosyncratic volatility components, and the Markov classification paths for unclassified variables. I will use the implementation details of Davidson, Hou and Koop (2025) as the baseline and adapt the code to the Taiwan dataset. Because the baseline system contains 52 variables, the estimation will rely on strong Bayesian shrinkage priors (such as Minnesota-type or Horseshoe priors adapted for SV models) to regularize the high-dimensional VAR coefficients, aggressively compress the parameter space, and limit overfitting while preserving the large information set needed to reduce omitted-variable bias. For the unclassified variables, the latent macro-financial classification indicators will be treated as finite-state Markov paths and sampled with a forward-filtering/backward-sampling step, so that classification probabilities reflect the full posterior path uncertainty rather than a sequence of

period-by-period labels.

The main posterior objects are the following:

$$\{h_{m,t}, h_{f,t}\}_{t=1}^T, \quad \{\pi_{i,t}\}_{i \in \mathcal{U}, t=1}^T,$$

where $\mathcal{U}$ denotes the set of unclassified variables. The first object gives Taiwan’s macroeconomic and financial uncertainty factors. The second object gives the time-varying classification probabilities for external and ambiguous domestic variables.

The first-year analysis will report four sets of results. First, I will plot Taiwan’s macroeconomic and financial uncertainty factors and compare their movements across major episodes. Second, I will plot classification probabilities for key external variables, including U.S. monetary indicators, Chinese real indicators, global financial risk, geopolitical risk, and trade-policy uncertainty. Third, I will compute impulse responses to macroeconomic and financial uncertainty shocks. Fourth, I will compute forecast error variance decompositions to quantify how much U.S., Chinese, and global uncertainty shocks contribute to Taiwan’s domestic uncertainty factors and observed economic variables.

The analysis will also include robustness checks. I will compare the full 52-variable model with a reduced benchmark specification that keeps the core Taiwan anchors and the most important external drivers. I will check alternative lag lengths, alternative sample starts, and specifications that exclude interpolated variables. For equity-market variables, I will compare the baseline TWSE total-return index with a longer price-index-based robustness sample when appropriate.

3.6 First-Year Work Plan and Anticipated Results

The first year is organized into four stages.

Period	Main task	Expected output
Months 1–4	Finalize the estimation dataset from the completed raw-data pipeline	Balanced panel, codebook, transformation log, stationarity checks, endpoint-gap decisions
Months 3–7	Adapt and test the OI-SVMVAR estimation code	Validated MCMC workflow and convergence diagnostics
Months 5–9	Estimate benchmark and full models	Posterior draws of $h_{m,t}$, $h_{f,t}$, and classification paths
Months 9–12	Analyze transmission channels and uncertainty effects	Classification plots, impulse responses, variance decompositions, and conference-paper draft

By the end of the first year, I expect to complete four outputs. First, I will produce a documented monthly dataset for Taiwan and external uncertainty drivers. Second, I will estimate Taiwan’s macroeconomic and financial uncertainty factors using the OI-SVMVAR framework. Third, I will produce time-varying classification probabilities for U.S., Chinese, global, and trade-policy variables. Fourth, I will prepare a first empirical paper draft or conference paper that documents the main first-year findings.

The first-year results will directly feed into the third year. The posterior draws of $h_{m,t}$ and $h_{f,t}$ will be passed into the BCA wedge validation exercise, so the project can evaluate structural mechanisms while accounting for the uncertainty generated in the first-stage estimation.

4 The Second Year: Methods, Procedures, and Implementation Schedule

4.1 Purpose of the Second Year

The first-year OI-SVMVAR identifies Taiwan’s macroeconomic and financial uncertainty factors and shows whether external drivers transmit mainly through macroeconomic or financial channels. However, this empirical evidence alone does not identify the underlying structural friction. A financial-channel response, for example, may reflect a collateral constraint, a time-varying risk premium, a working-capital requirement, or other financial frictions. The purpose of the second year is to build a theoretical bridge between the empirical uncertainty factors and structural economic mechanisms.

The second-year component develops a wedge-projection framework. The framework asks how second-moment uncertainty terms in a nonlinear structural model appear to an econometrician who observes the data through a first-order Business Cycle Accounting (BCA) prototype. The key idea is that uncertainty shocks may not appear as conventional first-order shocks in the true structural model, but their risk corrections can be projected into first-order BCA wedges. These wedges then provide diagnostic signatures for different transmission mechanisms.

The baseline nonlinear environment follows the small-open-economy framework with stochastic volatility developed by Fernández-Villaverde, Guerrón-Quintana, and co-authors (Fernández-Villaverde et al., 2011; Fernández-Villaverde and Guerrón-Quintana, 2020). In that environment, uncertainty affects real allocations through intertemporal collateral and risk-premium terms: tighter collateral risk, precautionary behavior, and time-varying risk premia distort the Euler equation and should therefore be visible primarily as an investment-wedge signature in a first-order BCA representation. I will contrast this baseline with a working-capital counter-hypothesis in which firms finance wage payments before production, so that higher external finance costs operate mainly through the static labor-demand margin and appear as an after-tax labor-wedge deterioration rather than an investment-wedge response.

This step is essential for the three-year project. Without it, the first-year evidence would identify when external uncertainty affects Taiwan and through which reduced-form channel, but it would not tell us which friction drives the response. The second year therefore supplies the theoretical mapping needed for the third-year empirical validation.

4.2 Second-Moment Wedge Projection

The central theoretical object is the BCA investment wedge. In a first-order BCA prototype, the investment wedge absorbs distortions that prevent the representative-household Euler equation from matching the observed path of consumption, investment, and capital returns. If the true data-generating process is a nonlinear structural model with stochastic volatility, the true Euler equation contains second-order risk corrections. The BCA prototype, which is first order, cannot

represent these risk corrections directly. Instead, it projects them into wedges. Appendix B states the economic model behind this projection, including the BCA Euler equation, the second-order risk correction, the closed-form projection, and the working-capital counter-hypothesis.

The preliminary theoretical work has derived the following projection equation:

$$\hat{\tau}_{x,t} = \omega_\tau E_t \hat{\tau}_{x,t+1} + \frac{1}{2} \Phi(\Omega_t),$$

where $\hat{\tau}_{x,t}$ is the log-deviation of the BCA investment wedge, ω_τ is a steady-state coefficient, Ω_t denotes the stochastic-volatility state, and $\Phi(\Omega_t)$ is the second-order risk correction in the true structural Euler equation.

This equation has a simple interpretation. A first-order BCA observer does not see the nonlinear risk correction directly. Instead, the observer recovers a forward-looking investment wedge that moves with the volatility state. If the volatility state follows an AR(1) process, the projection has a closed-form solution:

$$\hat{\tau}_{x,t} = \frac{\Phi(\bar{\Omega})}{2(1 - \omega_\tau)} + \frac{\phi}{2(1 - \omega_\tau \rho_\Omega)} (\Omega_t - \bar{\Omega}),$$

where $\phi = \Phi'(\bar{\Omega})$ and ρ_Ω is the persistence of the volatility state. This expression shows that persistent volatility can create a persistent first-order investment wedge even when the underlying mechanism operates through second-moment risk corrections.

The second-year task is to formalize this projection result, standardize the notation, and adapt the theorem to the small-open-economy setting relevant for Taiwan. The theorem provides the conceptual foundation for linking the first-year uncertainty factors to third-year BCA wedges.

4.3 Competing Mechanisms and Wedge Signatures

The wedge-projection framework is useful because different structural frictions imply different wedge signatures. The second year focuses on two benchmark mechanisms: an intertemporal collateral or risk-premium channel and a static working-capital channel. It also treats efficiency and external wedges as Taiwan-specific diagnostic objects because imported intermediate inputs, trade flows, and exchange-rate movements are central to a small open economy.

The first mechanism is a collateral or risk-premium channel. In this environment, higher financial uncertainty affects the intertemporal Euler equation through second-order risk corrections. These corrections may arise from precautionary motives, time-varying risk premia, collateral-risk terms, or aggregation effects in a two-agent economy. Because the distortion enters the intertemporal condition, the BCA projection appears mainly in the investment wedge. Under the baseline collateral-risk mechanism, financial uncertainty is expected to raise the investment wedge.

The second mechanism is a working-capital channel. In this environment, firms must finance part of their wage bill before production. A rise in external finance spreads then acts like a tax on labor demand. This mechanism does not primarily distort the intertemporal Euler equation. Instead, it distorts the static labor-demand condition. Therefore, the BCA projection appears mainly in the after-tax labor wedge. When financing spreads rise, the after-tax labor wedge should deteriorate. For Taiwan, however, working capital may also finance imported intermediate inputs rather than only wage payments. In that case, the same financing friction may also appear in the efficiency wedge or the government/external wedge, because the distortion affects materials, imported inputs, and external absorption. The empirical test will therefore treat the labor wedge as

the benchmark signature of a wage-finance channel while using the efficiency and external wedges as diagnostic objects for input-finance and trade-related variants of working-capital frictions.

The main wedge signatures are summarized as follows:

Mechanism	Main economic friction	Predicted BCA signature
Collateral or risk-premium channel	Intertemporal Euler distortion from uncertainty risk corrections	Investment wedge response
Working-capital channel	Static labor-demand distortion from financing costs	After-tax labor wedge response
SOE input-finance or trade variant	Imported-input finance, exchange-rate movements, or external absorption	Efficiency or external-wedge response

This table is the theoretical bridge between the first-year OI-SVMVAR and the third-year BCA validation. If the empirical uncertainty factors mainly move the investment wedge, the evidence supports an intertemporal collateral or risk-premium mechanism. If they mainly move the after-tax labor wedge, the evidence supports a working-capital mechanism. If they mainly move the efficiency or external wedge, the evidence suggests that Taiwan's small-open-economy structure, imported-input dependence, or external absorption margin is central to the transmission process. If several wedges respond, the data may indicate that several frictions operate at the same time.

4.4 Third-Order Extension

The second-year work also extends the projection framework to third order. This extension is needed because many stochastic-volatility DSGE models use third-order perturbation to capture the independent role of uncertainty shocks. At second order, the volatility state affects risk corrections, but the innovation to volatility may not generate the full impact response observed in generalized impulse responses. Third-order terms introduce additional objects that are empirically relevant.

The preliminary third-order extension introduces three new components. The first is a vol-of-vol correction, which refines the linear coefficient on the volatility state. The second is a state-dependent level-volatility interaction term, which allows the effect of uncertainty to depend on the current state of the economy. The third is an asymmetry term, which allows positive and negative volatility innovations to have responses that are not mirror images of one another.

These objects are useful for empirical interpretation. If the investment-wedge response is stronger during periods of tight credit conditions, this pattern would be consistent with a state-dependent collateral mechanism. If positive uncertainty shocks generate larger responses than negative uncertainty shocks, this asymmetry would support a nonlinear volatility mechanism. These predictions give the third-year empirical analysis more diagnostic power than a simple comparison of average impulse responses.

The second-year task is not to make the theoretical model unnecessarily complex. Instead, it is to derive a disciplined hierarchy of tests. The second-order projection provides the baseline wedge signature. The third-order extension provides additional checks for state dependence, asymmetry, and volatility-of-volatility effects.

4.5 Numerical Validation and Remaining Tasks

The preliminary numerical work has already identified an important issue. The closed-form projection coefficient is a local object. It should be compared with the impact response or the full horizon profile of the empirical wedge response, not mechanically with the maximum response at a long horizon. A peak impulse response may occur many months after the initial shock and may reflect persistence, propagation, and other state dynamics. Therefore, the second-year validation will compare the theoretical and empirical objects more carefully.

The numerical validation will proceed in four steps. First, I will compute the theoretical second-order wedge coefficients under the benchmark collateral and working-capital mechanisms. Second, I will simulate the structural models and recover the implied BCA wedges from the simulated data. Third, I will compare theoretical signs with simulated wedge responses at impact, at the zero crossing, and at the peak horizon. Fourth, I will evaluate the third-order predictions by studying state-dependent responses and asymmetric responses to positive and negative uncertainty innovations.

This validation strategy turns the theory into a practical empirical tool. It also reduces the risk that the third-year mechanism comparison relies on an incorrect theoretical object.

4.6 Second-Year Work Plan and Anticipated Results

The second year is organized into four stages.

Period	Main task	Expected output
Months 1–3	Finalize the second-moment wedge-projection theorem	Formal theorem, proof, and notation-consistent derivation
Months 3–6	Derive wedge signatures for benchmark mechanisms	Investment-wedge and labor-wedge diagnostic table
Months 6–9	Develop and refine the third-order extension	State-dependence, asymmetry, and vol-of-vol predictions
Months 9–12	Conduct numerical validation and prepare theory paper	Validated theoretical appendix and standalone theory manuscript

By the end of the second year, I expect to complete four outputs. First, I will produce a formal wedge-projection theorem that maps stochastic-volatility risk corrections into BCA wedges. Second, I will derive a wedge-signature table that distinguishes collateral, risk-premium, and working-capital mechanisms. Third, I will complete numerical validation using simulated structural models. Fourth, I will prepare a theoretical paper or appendix that can support the third-year empirical validation and eventual journal submission.

The main contribution of the second year is therefore not only theoretical. It provides the identification logic that allows the full project to move from reduced-form uncertainty measurement to structural economic interpretation.

5 The Third Year: Methods, Procedures, and Implementation Schedule

5.1 Purpose of the Third Year

The third year empirically validates the structural mechanisms derived in the second year using Taiwan's aggregate data. The first-year OI-SVMVAR identifies external uncertainty transmission and extracts Taiwan's macroeconomic and financial uncertainty factors. The second-year wedge-projection theory derives the BCA signatures implied by competing mechanisms. The third year connects these two components by testing whether Taiwan's empirical wedges respond to uncertainty shocks in the way predicted by the theory.

This step is necessary because reduced-form evidence alone cannot distinguish structural mechanisms. If financial uncertainty raises asset-market volatility and depresses investment, the response may reflect a collateral constraint, a risk-premium channel, a working-capital channel, or a combination of frictions. The BCA framework provides a way to separate these mechanisms because different frictions project into different wedges. The third year therefore asks whether Taiwan's observed wedge responses are closer to the investment-wedge signature of the collateral or risk-premium channel, or to the labor-wedge signature of the working-capital channel.

5.2 Constructing Taiwan's BCA Wedges

The first task in the third year is to construct a small-open-economy BCA prototype for Taiwan. The prototype decomposes Taiwan's aggregate fluctuations into four accounting wedges: an efficiency wedge, a labor wedge, an investment wedge, and a government or external wedge. These wedges summarize the distortions that a representative-agent prototype must introduce in order to reproduce the observed paths of output, consumption, investment, labor input, and external absorption.

The empirical construction will use Taiwan aggregate data from national accounts, labor-market statistics, price indices, and external-sector accounts. The baseline BCA exercise will be conducted at quarterly frequency because Taiwan's core national-account objects, including GDP, private consumption, fixed capital formation, and external absorption, are measured most reliably at that frequency. The monthly uncertainty factors from the first-year OI-SVMVAR will therefore be time-aggregated to quarterly frequency before entering the wedge-level GIRF computations. The baseline aggregation will use quarterly averages of the monthly posterior draws, with end-of-quarter values used as a robustness check for shocks that are more closely tied to financial-market conditions. This design preserves the first-year model's ability to detect short-lived uncertainty episodes while keeping the third-year BCA wedges anchored in reliable national-account data.

The main empirical focus is on two wedges:

$$\hat{\tau}_{x,t} \quad \text{and} \quad \hat{\Lambda}_{\ell,t}.$$

The first object is the investment wedge. It captures distortions in the intertemporal Euler equation and is the key empirical object for collateral and risk-premium mechanisms. The second object is the after-tax labor wedge. It captures distortions in the intratemporal labor condition and is the key empirical object for working-capital mechanisms.

The remaining BCA wedges are also informative. The efficiency wedge helps distinguish uncertainty effects from productivity-like movements. The government or external wedge helps absorb changes in public spending, trade balances, and external absorption. These wedges will be reported to ensure that the mechanism comparison is not driven by omitted accounting components.

5.3 Linking Uncertainty Factors to Wedge Signatures

The core third-year exercise constructs an empirical wedge IRF profile from the first-year OI-SVMVAR posterior draws and compares it against theoretically simulated wedge paths from competing structural models. The two-step BCA accounting procedure is applied identically on the empirical and theoretical sides, ensuring that the RMSE comparison is not contaminated by differences in the measurement protocol.

Empirical wedge IRF construction. Let $\varepsilon_{m,t}$ and $\varepsilon_{f,t}$ denote the macroeconomic and financial uncertainty shocks identified in the first-year OI-SVMVAR. For each posterior draw $\theta^{(s)}$, the model's impulse response functions yield draw-specific paths of output $\hat{y}_{t+h}^{(s)}$, consumption $\hat{c}_{t+h}^{(s)}$, investment $\hat{x}_{t+h}^{(s)}$, and labor $\hat{\ell}_{t+h}^{(s)}$ in response to a shock of size δ . The BCA prototype accounting identities are then applied to each draw to recover the corresponding wedge responses:

$$\hat{A}_h^{(s)} = \hat{y}_h^{(s)} - \alpha \hat{k}_{h-1}^{(s)} - (1 - \alpha) \hat{\ell}_h^{(s)}, \quad (1)$$

$$\widehat{(1 - \tau_\ell)}_h^{(s)} = (\nu + 1) \hat{\ell}_h^{(s)} + \gamma \hat{c}_h^{(s)} - \hat{y}_h^{(s)}, \quad (2)$$

$$\widehat{(1 + \tau_x)}_h^{(s)} = \text{backward Euler induction from } \hat{c}, \hat{y}, \hat{k}, \quad (3)$$

where (1)–(3) are the standard CKM log-linearized accounting conditions. The posterior-averaged empirical wedge IRF profile is

$$\hat{w}_h^{\text{emp}}(\delta) = \frac{1}{S} \sum_{s=1}^S \hat{w}_h^{(s)}(\delta),$$

with credible bands constructed from the 16th and 84th posterior quantiles across draws.

The theoretical predictions from the second year guide the interpretation. If financial uncertainty mainly raises the investment wedge, the evidence supports a collateral or risk-premium mechanism. If financial uncertainty mainly lowers the after-tax labor wedge, the evidence supports a working-capital mechanism. If macroeconomic uncertainty affects the efficiency wedge or the external wedge more strongly, the evidence may point to real-side trade, productivity, or external absorption channels. This broader wedge comparison is important for Taiwan. If working-capital frictions operate through imported materials or intermediate inputs rather than only through wage payments, the response may appear partly in the efficiency wedge or the external wedge. I will therefore report the full wedge vector and interpret labor-wedge evidence as the benchmark wage-finance case, while treating efficiency- and external-wedge responses as evidence for input-finance or trade-finance variants of the working-capital channel.

This wedge-level approach is more informative than matching impulse responses of standard macroeconomic variables alone. A decline in investment, for example, can arise from many mechanisms. The investment wedge asks whether the decline reflects an intertemporal distortion. The labor wedge asks whether the shock acts like a distortion in the labor-demand or labor-supply condition. The efficiency and external wedges help assess whether the response instead resembles

productivity or external-demand movements.

5.4 MCMC Uncertainty Propagation

A common limitation of two-step empirical designs is generated-regressor bias. If the uncertainty factor is estimated in the first step and then treated as observed in the second step, the second-step confidence intervals may be too narrow. This issue is especially relevant here because $h_{m,t}$ and $h_{f,t}$ are latent factors estimated from a large Bayesian model.

To address this problem, the third-year analysis propagates first-year posterior uncertainty directly into the empirical wedge IRF construction. Rather than conditioning on a single posterior mean draw, I retain the full set of posterior draws $\{\theta^{(s)}\}_{s=1}^S$ from the first-year OI-SVMVAR and apply the BCA accounting identities (1)–(3) to each draw’s IRF path independently. The final credible intervals therefore integrate over both the uncertainty in the first-year shock identification and the draw-to-draw variation in the resulting wedge responses.

This procedure has two advantages. First, it keeps the empirical validation fully consistent with the Bayesian structure of the first-year model, so that the three years of the project share a coherent posterior framework. Second, it makes the mechanism comparison more conservative. If the data strongly support one mechanism even after posterior uncertainty is propagated, the conclusion will be more credible. If the posterior uncertainty renders the evidence ambiguous, the project will report the relative strength of evidence rather than forcing a binary classification.

The procedure is also computationally feasible. The expensive step is the first-year OI-SVMVAR estimation. Once posterior draws are retained, applying the BCA accounting identities to each draw’s IRF path involves only closed-form matrix operations and a backward recursion for the investment wedge, which is far less demanding than re-estimating the full high-dimensional Bayesian VAR.

5.5 Hypothesis Testing and Policy Interpretation

The third-year mechanism comparison is organized as a quantitative model selection exercise. Rather than asking whether a regression coefficient has the right sign, the exercise asks which structural model best reproduces Taiwan’s empirically observed wedge responses. The model whose theoretically predicted wedge path most closely matches the GIRF profile estimated from the data is selected as the dominant channel. This framing imposes a strict and objective standard that is directly traceable to the first-year OI-SVMVAR output.

The procedure follows four steps.

Step 1 — Empirical target. Using the macroeconomic and financial uncertainty shocks $\varepsilon_{m,t}$ and $\varepsilon_{f,t}$ identified by the first-year OI-SVMVAR, I extract the posterior IRF paths of output, consumption, investment, and labor for each of the S posterior draws. The BCA accounting identities (1)–(3) are applied to each draw to recover the corresponding wedge IRF paths. The posterior-averaged empirical wedge profile $\hat{w}_h^{\text{emp}}(\delta)$, together with its 68% credible bands, constitutes the empirical target.

Step 2 — Theoretical candidates. The second-year theory produces wedge-projection signatures for two competing structural mechanisms: the collateral and risk-premium channel ($\mathcal{M}^{\text{coll}}$, FVGQ-style model) and the working-capital channel ($\mathcal{M}^{\text{wc}}$). Under each model, I simulate structural dynamics in response to an uncertainty shock of size δ .

Step 3 — Simulated wedge extraction. For each structural model candidate, I compute the Generalized Impulse Response Function (Koop, Pesaran and Potter, 1996) of the BCA wedges via an antithetic-pair simulation design. Starting from the model’s stochastic steady state (SSS), N matched path pairs of length H are generated. Within each pair, a baseline path sets the volatility shock to zero ($\varepsilon_t = 0$) while the shocked path sets it to $\varepsilon_t = \delta$; all remaining innovations are shared. The BCA accounting identities (1)–(3) are applied to every simulated path to extract the wedge series. The theoretical GIRF is the cross-pair mean difference:

$$\hat{w}_h^m = \frac{1}{N} \sum_{i=1}^N \left[w_h^{m,\text{shock},i} - w_h^{m,\text{base},i} \right],$$

with 5th–95th cross-pair quantile bands providing a measure of simulation uncertainty.

Step 4 — Quantitative matching. The fit of each structural model is evaluated by the root mean squared error between the empirical wedge IRF profile and the theoretical wedge GIRF:

$$\text{RMSE}^m = \left(\frac{1}{H+1} \sum_{h=0}^H \left[\hat{w}_h^{\text{emp}}(\delta) - \hat{w}_h^m \right]^2 \right)^{1/2}.$$

The dominant mechanism is the model with the smallest RMSE. A correlation-based criterion serves as a robustness check.

The formal model-selection hypotheses are:

$$H_0^{\text{coll}} : \quad \text{RMSE}^{\text{coll}} < \text{RMSE}^{\text{wc}} \quad \text{for the investment-wedge GIRF path,}$$

and

$$H_0^{\text{wc}} : \quad \text{RMSE}^{\text{wc}} < \text{RMSE}^{\text{coll}} \quad \text{for the labor-wedge GIRF path.}$$

The first hypothesis identifies the collateral or risk-premium mechanism as dominant through the intertemporal Euler equation. The second identifies the working-capital mechanism as dominant through the intratemporal labor condition. A clear RMSE gap across models constitutes strong evidence for a single mechanism; overlapping posterior credible intervals indicate that the data support a mixed or time-varying channel.

The policy implications differ sharply across mechanisms. If H_0^{coll} is supported, external uncertainty primarily affects Taiwan by tightening intertemporal financing conditions, and policy attention should focus on credit-market stabilization, liquidity provision, and financial-market resilience. If H_0^{wc} is supported, external uncertainty acts more directly through firms’ short-term financing costs and labor demand, and policies that reduce working-capital pressure on firms become more relevant. If real-side wedges achieve the smallest RMSE instead, the appropriate response may involve export support, industrial adjustment, or supply-chain resilience.

The empirical goal is not to assign all episodes to a single permanent mechanism. Taiwan’s transmission channel may change over time. The Global Financial Crisis may display a stronger collateral-channel component while the U.S.–China trade war may show stronger real and external-demand components. The third-year analysis will therefore report both the full-sample RMSE comparison and episode-specific wedge IRF evidence when the data allow.

5.6 Third-Year Work Plan and Anticipated Results

The third year is organized into four stages.

Period	Main task	Expected output
Months 1–3	Construct Taiwan BCA prototype and wedge series	Efficiency, labor, investment, and government/external wedges
Months 3–6	Link Year 1 uncertainty posterior draws to wedge data	Matched uncertainty-wedge dataset and posterior shock measures
Months 6–9	Compute MCMC-integrated GIRF paths and theoretical wedge simulations	Empirical GIRF profiles and structural-model wedge candidates
Months 9–12	Test structural mechanisms and write final manuscript	Mechanism comparison, policy interpretation, and journal-paper draft

By the end of the third year, I expect to complete four outputs. First, I will produce a Taiwan BCA wedge dataset suitable for studying uncertainty transmission. Second, I will compute MCMC-integrated GIRF paths for Taiwan’s investment and labor wedges in response to the macroeconomic and financial uncertainty shocks identified in the first year. Third, I will conduct a formal model selection exercise using RMSE comparison between the empirical GIRF profiles and the theoretically simulated wedge paths derived in the second year. Fourth, I will prepare a complete manuscript that integrates the first-year OI-SVMVAR evidence, the second-year wedge-projection theory, and the third-year Taiwan BCA validation.

The third year completes the project’s main contribution. It transforms the first-year uncertainty measurement exercise and the second-year theoretical mapping into direct evidence on the structural mechanism through which external uncertainty affects Taiwan.

6 Integrated Research Project

6.1 Integration Across the Three Years

This project is designed as a unified three-year research program rather than three separate studies. The first year identifies where external uncertainty comes from and whether it enters Taiwan mainly through macroeconomic or financial channels. The second year turns those empirical channels into structural predictions by deriving wedge signatures for collateral, risk-premium, and working-capital mechanisms. The third year then tests these signatures in Taiwan’s BCA wedges while propagating posterior uncertainty from the first-year OI-SVMVAR.

The integration matters for policy because each year removes one layer of ambiguity. A finding that U.S. monetary uncertainty raises Taiwan’s financial uncertainty is informative, but it does not by itself say whether the relevant policy margin is liquidity, collateral, credit access, or real-side adjustment. The wedge-projection and BCA validation stages translate the empirical classification evidence into mechanism-level diagnoses, so that the final project can distinguish which policy instruments are most relevant under different external uncertainty episodes.

6.2 Expected Academic Contributions

The project makes four academic contributions.

First, it provides a new empirical account of external uncertainty transmission to Taiwan. Existing work often studies uncertainty using a single proxy, such as financial volatility or economic policy uncertainty. This project instead estimates domestic uncertainty factors jointly with their macroeconomic effects and external drivers. This design allows the project to compare U.S., Chinese, global, and trade-policy sources within one coherent framework.

Second, the project adapts the OI-SVMVAR framework to a small open economy. The key innovation is to reinterpret the unclassified-variable mechanism as a transmission-channel identifier. Instead of pre-assigning external variables to macroeconomic or financial blocks, the model lets the data determine whether each external driver is associated with Taiwan's macroeconomic or financial uncertainty at each point in time. This approach provides a template for studying other small open economies exposed to multiple external shocks.

Third, the project develops a wedge-projection framework for interpreting uncertainty shocks. The framework shows how second-moment uncertainty corrections in nonlinear structural models can appear as first-order BCA wedges. This contribution helps bridge reduced-form uncertainty measurement and micro-founded structural interpretation.

Fourth, the project provides an empirical mechanism test using Taiwan's BCA wedges. Instead of comparing only the impulse responses of standard macroeconomic variables, the project tests whether uncertainty shocks move the wedges predicted by theory. This wedge-level validation offers a sharper way to distinguish collateral, risk-premium, and working-capital channels.

6.3 Expected Policy Contributions

The project also has direct policy relevance for Taiwan. External uncertainty is a recurring feature of Taiwan's economic environment. U.S. monetary tightening, China-related demand shocks, trade-policy disputes, global financial volatility, and geopolitical risk all affect Taiwan, but they do not necessarily operate through the same channel. A policy response that is appropriate for a real-side demand shock may be less effective for a financial-market shock.

By distinguishing macroeconomic and financial transmission channels, the project can help policymakers interpret uncertainty episodes more precisely. If an external shock mainly affects Taiwan through macroeconomic channels, the relevant policy tools are closer to MOEA and NDC instruments, including export credit guarantees, small and medium enterprise (SME) liquidity support, supply-chain resilience programs, and industrial adjustment measures. If the shock mainly affects Taiwan through financial channels, the relevant tools are closer to CBC liquidity operations, macroprudential measures, selective credit controls, exchange-market monitoring, and Financial Supervisory Commission (FSC) monitoring of market functioning and financial-stability risks.

The third-year wedge validation further improves the policy interpretation. If the evidence points to a collateral or risk-premium channel, CBC and FSC financial-stability tools become more central. If the evidence points to a working-capital channel, SME liquidity facilities, export credit guarantees, and bank-credit support become more relevant. If real-side wedges dominate, MOEA and NDC export promotion, supply-chain adjustment, and industrial upgrading policies may be more appropriate.

The project therefore contributes not only to academic research on uncertainty shocks, but also

to the practical diagnosis of Taiwan’s external vulnerability.

6.4 Expected Outputs and Deliverables

The expected outputs are organized by year.

Year	Main research output	Deliverables
Year 1	Empirical identification of Taiwan’s uncertainty transmission	Monthly dataset, OI-SVMVAR estimates, uncertainty factors, classification probabilities, IRFs, FEVDs, conference-paper draft
Year 2	Theoretical wedge-projection framework	Formal theorem, wedge-signature table, numerical validation, theoretical paper or appendix
Year 3	Empirical BCA validation and mechanism testing	Taiwan BCA wedge dataset, MCMC-integrated GIRF profiles, RMSE-based mechanism comparison, journal-paper draft

The final project will produce three main research products. The first is a Taiwan external uncertainty database and empirical paper. The second is a theoretical wedge-projection paper or appendix. The third is an integrated empirical-structural paper on Taiwan’s external uncertainty transmission.

6.5 Risk Management: Anticipated Problems and Means of Resolution

The project is ambitious, but its risks are manageable. The main risks and mitigation strategies are summarized below.

Risk	Mitigation strategy
The full 52-variable OI-SVMVAR is slow to converge.	Begin with a systematically reduced benchmark model (e.g., a 15-to-20 variable system) focusing on the core Taiwan macroeconomic and financial anchors and the most prominent external indices. Move to the full high-dimensional model incrementally after validating the sampler and convergence diagnostics.
Some data series require interpolation or have limited coverage.	Report the baseline specification transparently, maintain a complete data-construction log, and conduct robustness checks that exclude interpolated variables or use alternative sample windows.
China-related monthly data are difficult to harmonize.	Use the downloaded China industrial production, PPI, and M2 series after source validation, and compare them with alternative public sources when available. If definitions or vintages remain fragile, report a reduced China block as a robustness specification while keeping U.S. and global indicators in the baseline.
Theoretical wedge signatures are not sharply separated in all cases.	Report the relative strength of evidence across mechanisms rather than forcing a binary classification. Use the timing, sign, and wedge location of the responses jointly.
Wedge impulse response profiles can be sensitive to horizon choice and identification assumptions.	Report full horizon profiles, including impact responses, peak responses, and sign changes, and compare against the theoretical generalized impulse response across the full horizon rather than at a single long-horizon peak.
First-stage uncertainty factors create generated-regressor concerns.	Propagate posterior uncertainty from the first-year OI-SVMVAR directly into the third-year wedge impulse response construction, with each posterior draw producing its own BCA-wedge response path, instead of conditioning on posterior-mean factors.

These safeguards ensure that the project remains feasible even if some components require adjustment. The reduced benchmark model protects the first-year empirical analysis. The theoretical wedge-signature table protects the second-year mechanism comparison. The MCMC-integrated validation protects the third-year inference. Together, these safeguards make the project suitable for a three-year research grant.

6.6 Concluding Statement

This project studies Taiwan as a small open economy caught between major external sources of uncertainty. Its contribution is to move beyond asking whether uncertainty matters. Instead, it asks where uncertainty comes from, how it enters Taiwan, and which structural friction explains its effects.

The first year provides empirical identification through an order-invariant large-scale SVMVAR. The second year provides theoretical interpretation through wedge projection. The third

year provides empirical validation through Taiwan's BCA wedges. By combining these three components, the project delivers a coherent framework for measuring, interpreting, and validating external uncertainty transmission in Taiwan.

A Econometric Model: OI-SVMVAR

This appendix summarizes the order-invariant stochastic volatility in mean vector autoregression (OI-SVMVAR) used in the first-year empirical analysis. Let $\mathbf{y}_t$ denote the $N \times 1$ vector of transformed and standardized monthly observables, partitioned into macroeconomic, unclassified, and financial blocks. The observation equation is

$$\mathbf{B}_0 \mathbf{y}_t = \sum_{\ell=1}^p \mathbf{B}_\ell \mathbf{y}_{t-\ell} + \sum_{j=0}^q \mathbf{A}_j^h \mathbf{h}_{t-j} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim N(\mathbf{0}, \mathbf{U}_t), \quad (4)$$

where $\mathbf{h}_t = (h_{m,t}, h_{f,t})'$ contains Taiwan's common macroeconomic and financial log-volatility factors. The matrices $\mathbf{A}_j^h$ capture the stochastic-volatility-in-mean channel: changes in uncertainty can directly shift the conditional mean of output, employment, asset prices, credit conditions, and other observables. The contemporaneous impact matrix $\mathbf{B}_0$ is unrestricted apart from the normalization of its diagonal elements to one. This is the key order-invariance feature of the model: inference is not tied to a Cholesky ordering of the variables in $\mathbf{y}_t$.

The conditional variance matrix in the structural equation is diagonal and block-partitioned as

$$\mathbf{U}_t = \begin{pmatrix} \mathbf{U}_{m,t} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{U}_{u,t} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{U}_{f,t} \end{pmatrix}, \quad \mathbf{U}_{k,t} = \text{diag}(e^{\omega_{1,t}^k}, \dots, e^{\omega_{n_k,t}^k}), \quad k \in \{m, u, f\}. \quad (5)$$

The block ordering is not an identifying recursive ordering; it only records whether a variable is anchored as macroeconomic, left unclassified, or anchored as financial. Contemporaneous interactions among all variables remain unrestricted through $\mathbf{B}_0$.

The log-volatility of each observable is decomposed into an idiosyncratic component and one of the two common uncertainty factors:

$$\begin{aligned} \omega_{i,t}^m &= \eta_{i,t}^m + h_{m,t}, & i &= 1, \dots, n_m, \\ \omega_{i,t}^u &= \eta_{i,t}^u + h_{s_{i,t},t}, & s_{i,t} &\in \{m, f\}, \quad i = 1, \dots, n_u, \\ \omega_{i,t}^f &= \eta_{i,t}^f + h_{f,t}, & i &= 1, \dots, n_f. \end{aligned} \quad (6)$$

The idiosyncratic terms $\eta_{i,t}^k$ capture variable-specific volatility movements. Variables classified ex ante as macroeconomic load on $h_{m,t}$, and variables classified ex ante as financial load on $h_{f,t}$. For unclassified variables, including external U.S., Chinese, global, and policy-risk indicators, the latent state $s_{i,t}$ determines whether the variable's common volatility component is assigned to the macroeconomic or financial factor at time t .

The two common factors evolve jointly according to

$$\mathbf{h}_t = \sum_{r=1}^{p_h} \boldsymbol{\Phi}_r \mathbf{h}_{t-r} + \sum_{s=1}^{p_y} \boldsymbol{\Psi}_s \mathbf{y}_{t-s} + \boldsymbol{\varepsilon}_t^h, \quad \boldsymbol{\varepsilon}_t^h \sim N(\mathbf{0}, \boldsymbol{\Sigma}_h). \quad (7)$$

This specification allows macroeconomic and financial uncertainty to interact over time and allows lagged domestic and external conditions to forecast Taiwan’s uncertainty factors. The factors are therefore estimated jointly with the VAR rather than constructed in a preliminary step.

For each unclassified variable i , the state $s_{i,t}$ follows a two-state first-order Markov chain. The main transmission-channel object is the posterior classification probability

$$\pi_{i,t} = P(s_{i,t} = m \mid \mathbf{y}_{1:T}), \quad (8)$$

which is the posterior probability that variable i belongs to the macroeconomic block at time t . Values of $\pi_{i,t}$ near one indicate macroeconomic transmission, while values near zero indicate financial transmission. Because $\pi_{i,t}$ is time-varying, the same external driver can operate through different channels in different episodes.

The model is estimated by Bayesian MCMC following the OI-SVMVAR algorithm of Davidson, Hou and Koop (2025). The sampler cycles through the VAR and volatility-in-mean coefficients, the unrestricted contemporaneous matrix $\mathbf{B}_0$, the common and idiosyncratic log-volatility paths, and the latent classification paths. The unrestricted $\mathbf{B}_0$ is sampled using the order-invariant row-wise parameter transformation developed for the model, preserving permutation-invariant inference without imposing block exogeneity or recursive zero restrictions. The finite-state Markov paths $(s_{i,1}, \dots, s_{i,T})$ and their transition probabilities are drawn with forward-filtering backward-sampling and Chib-style path sampling. Posterior draws of $\mathbf{h}_t$ and $\pi_{i,t}$ are then used for the classification, variance-decomposition, and impulse-response analyses.

B Economic Model: Structural Environment and Wedge Projection

This appendix records the compact economic model behind the wedge-projection exercise. The purpose is not to estimate a full DSGE model inside the proposal, but to make clear which structural environment generates the wedge signatures used in the second and third years. The baseline is a Fernández-Villaverde–Guerrón-Quintana-style two-agent collateral economy with stochastic volatility in real, demand, and financial-friction shocks. I then contrast this intertemporal channel with a working-capital alternative.

Time is discrete. A household contains workers and entrepreneurs. The household head allocates equity claims and writes contingent plans before idiosyncratic member types are realized. Workers supply labor and receive wages; entrepreneurs invest in capital and face a collateral constraint. A compact representation of preferences is

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t d_t \left[\pi \frac{(c_t^E)^{1-\rho} - 1}{1-\rho} + (1-\pi) \frac{\{c_t^W (1-\ell_t)^\eta\}^{1-\rho} - 1}{1-\rho} \right], \quad (9)$$

where c_t^E and c_t^W are entrepreneur and worker consumption, ℓ_t is labor, d_t is a preference shifter, and π is the fraction of entrepreneurs.

Final output is produced competitively using utilized capital and labor:

$$y_t = A_t (u_t k_t)^\alpha \ell_t^{1-\alpha}, \quad r_t = \alpha \frac{y_t}{u_t k_t}, \quad w_t = (1-\alpha) \frac{y_t}{\ell_t}. \quad (10)$$

Capital utilization u_t raises depreciation through a convex schedule

$$\delta(u_t) = \delta_0 + \delta_1(u_t - 1) + \frac{\delta_2}{2}(u_t - 1)^2, \quad k_{t+1} = (1 - \delta(u_t))k_t + \pi i_t. \quad (11)$$

The key financial friction is a skin-in-the-game collateral constraint. If s_{t+1}^E denotes entrepreneur equity claims carried into the next period, then

$$s_{t+1}^E \geq (1 - \theta_t)i_t + (1 - \phi)(1 - \delta(u_t))s_t. \quad (12)$$

The time-varying pledgeability parameter θ_t controls how much new capital can be externally financed, while ϕ controls resaleability of existing capital. When the constraint binds, changes in uncertainty alter the intertemporal Euler equation through collateral-risk, risk-premium, and precautionary terms.

The structural shocks are a real productivity shock A_t , a demand shifter d_t , and a financial friction shock θ_t . Each follows a persistent level process with stochastic volatility:

$$\begin{aligned} z_{j,t} &= (1 - \rho_j)\bar{z}_j + \rho_j z_{j,t-1} + \exp(\sigma_{j,t})\varepsilon_{j,t}, & \varepsilon_{j,t} &\sim N(0, 1), \\ \sigma_{j,t} &= (1 - \rho_{\sigma_j})\bar{\sigma}_j + \rho_{\sigma_j}\sigma_{j,t-1} + (1 - \rho_{\sigma_j}^2)^{1/2}\nu_j u_{j,t}, & u_{j,t} &\sim N(0, 1), \end{aligned} \quad (13)$$

for $j \in \{A, d, \theta\}$. The volatility innovations $u_{j,t}$ are the structural uncertainty shocks. In the empirical project, the OI-SVMVAR uncertainty factors provide observed counterparts to these latent volatility states.

In a representative-household BCA prototype, the log-linearized investment Euler equation can be written as

$$\hat{\tau}_{x,t} - \gamma \tilde{C}_t = \mathbb{E}_t \left[-\gamma \tilde{C}_{t+1} + \omega_K \tilde{R}_{t+1}^k + \omega_\tau \hat{\tau}_{x,t+1} \right], \quad (14)$$

where $\hat{\tau}_{x,t}$ is the investment wedge, $\tilde{C}_t$ is measured consumption, $\tilde{R}_{t+1}^k$ is the measured capital return, γ is the risk-aversion parameter, and ω_K, ω_τ are steady-state coefficients. If the true data-generating process contains stochastic volatility, the corresponding Euler equation has a second-order risk correction:

$$-\gamma \tilde{C}_t = \mathbb{E}_t \left[-\gamma \tilde{C}_{t+1} + \omega_K \tilde{R}_{t+1}^k \right] + \frac{1}{2} \Phi(\Omega_t) + o(\|\Delta\Omega_t\|), \quad (15)$$

where Ω_t denotes the volatility state and $\Phi(\Omega_t)$ collects conditional variance, covariance, and Hessian terms. Comparing (14) and (15) yields the wedge projection equation

$$\hat{\tau}_{x,t} = \omega_\tau \mathbb{E}_t \hat{\tau}_{x,t+1} + \frac{1}{2} \Phi(\Omega_t). \quad (16)$$

Let $\Delta\Omega_t = \Omega_t - \bar{\Omega}$ and suppose the volatility state follows a stable matrix AR(1),

$$\Delta\Omega_{t+1} = R_\Omega \Delta\Omega_t + \eta_{t+1}, \quad \rho(R_\Omega) < 1, \quad (17)$$

with local approximation $\Phi(\Omega_t) = \Phi(\bar{\Omega}) + \phi' \Delta\Omega_t + o(\|\Delta\Omega_t\|)$. If $|\omega_\tau| < 1$, the locally bounded solution is

$$\hat{\tau}_{x,t} = \frac{\Phi(\bar{\Omega})}{2(1 - \omega_\tau)} + \frac{1}{2} \phi' (I - \omega_\tau R_\Omega)^{-1} \Delta\Omega_t + o(\|\Delta\Omega_t\|). \quad (18)$$

For a scalar AR(1), $\Delta\Omega_{t+1} = \rho_\Omega\Delta\Omega_t + \eta_{t+1}$, this becomes

$$\hat{\tau}_{x,t} = \frac{\Phi(\bar{\Omega})}{2(1 - \omega_\tau)} + \underbrace{\frac{\phi}{2(1 - \omega_\tau\rho_\Omega)}}_{\kappa_x}(\Omega_t - \bar{\Omega}) + o(|\Omega_t - \bar{\Omega}|). \quad (19)$$

Thus a purely second-moment structural force may be recovered by BCA as a persistent first-order investment wedge.

In this collateral model, the risk correction can be organized schematically as

$$\Phi^{collat}(\Omega_t) = \Phi^{prec}(\Omega_t) + \Phi^{rp}(\Omega_t) + \Phi^{coll}(\Omega_t) + \Phi^{agg}(\Omega_t). \quad (20)$$

The precautionary term is increasing in conditional consumption risk, $\Phi_t^{prec} \propto \gamma^2 \text{Var}_t(\hat{c}_{t+1})$. The risk-premium term depends on the consumption-return covariance, $\Phi_t^{rp} \propto -\gamma \text{Cov}_t(\hat{c}_{t+1}, \hat{r}_{t+1}^k)$. The collateral term captures the effect of volatility on pledgeability or collateral tightness, $\Phi_t^{coll} \propto \nu_\theta \sigma_{\theta,t}^2$, where ν_θ governs the sensitivity of the collateral constraint to financial-friction volatility. In a two-agent version, aggregation adds a dispersion term, for example $\Phi_t^{agg} \propto \text{Var}_t(\hat{c}_{E,t+1} - \hat{c}_{W,t+1})$, because the Euler equation pricing investment is tied to entrepreneur marginal utility while BCA uses aggregate consumption. These components imply that the sign of κ_x is disciplined by the relative strength of precautionary motives, risk premia, collateral risk, and aggregation.

A working-capital mechanism has a different signature. Suppose firms finance a fraction θ_{wc} of the wage bill in advance and face external finance spread q_t . The true static labor condition is

$$A_t F_{L,t} = w_t(1 + \theta_{wc}q_t), \quad (21)$$

where $F_{L,t}$ is the marginal product of labor. The BCA intratemporal condition is

$$-\frac{U_L(C_t, L_t)}{U_C(C_t, L_t)} = \Lambda_{\ell,t}^{BCA} A_t F_{L,t}. \quad (22)$$

Here U_C and U_L denote the marginal utilities of consumption and labor. Using the household condition $-U_L/U_C = w_t$ gives the exact projection

$$\Lambda_{\ell,t}^{BCA} = \frac{1}{1 + \theta_{wc}q_t}. \quad (23)$$

Around $\bar{q}$,

$$\hat{\Lambda}_{\ell,t}^{BCA} = -\frac{\theta_{wc}}{1 + \theta_{wc}\bar{q}}(q_t - \bar{q}) + O((q_t - \bar{q})^2). \quad (24)$$

If $q_t - \bar{q} = \alpha_\xi \hat{\xi}_t + \alpha_\Omega(\Omega_t - \bar{\Omega}) + o(\cdot)$ with $\alpha_\xi > 0$ and $\alpha_\Omega \geq 0$, a financial tightening or higher uncertainty lowers the after-tax labor wedge. The model therefore distinguishes an intertemporal collateral/risk-premium channel, which projects mainly into $\hat{\tau}_{x,t}$, from a working-capital channel, which projects mainly into $\hat{\Lambda}_{\ell,t}^{BCA}$.

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